

Colin Farley

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EDUCATION

Rensselaer Polytechnic Institute

Bachelor of Science in Computer Science, GPA: 3.83

Troy, NY

Graduated: May 2025

EXPERIENCE

Quantum Machine Learning Fellow

May 2025 – August 2025

HARP Research, Inc.

Remote

- Conducted research on quantum annealing algorithm design for machine learning, focusing on applications in HPC, finance, and optimization using simulated bifurcation machines.
- Found and benchmarked **3 - 5x** speed-up in problem encoding for iterative quadratic unconstrained binary optimization (QUBO) problems.

AI Engineer

January 2025 – February 2025

Bland AI

Remote

- Prompt engineered AI phone agents for businesses using Bland's Development Platform.
- Integrated LLMs with APIs and variable extraction to automate full workflows.

Product Development Intern

June 2024 – August 2024

Crown Castle

Remote

- Designed PowerBI dashboards for CRM insights, reducing internal cost request wait times by 18%.
- Led product design and development of an internal CPQ tool using Next.js.

Undergraduate Teaching Assistant

Sept. 2022 – Jan. 2024

Rensselaer Polytechnic Institute

Troy, NY

- Mentored 30+ students and ran 2 labs per week on data structures, algorithms, and programming basics.

PROJECTS

Safe Commute

March 2025

- Won 1st place** at the Hudson Valley Tech Fest Hackathon
- Developed a **full-stack web app** that used public geospatial crime data and Google Maps to suggest the safest routes based on recent incidents
- Deployed a **working prototype in under 3 hours**, visualizing real-time crime data with interactive heatmaps and integrating route safety analysis into a seamless user interface.

Smart Traffic Signal Optimization using Deep Reinforcement Learning

February 2025 – Present

- Developing an adaptive traffic signal control system in SUMO, leveraging Deep Q-Networks or Proximal Policy Optimization to minimize urban congestion.
- Designing a tailored reward function that incentivizes shorter waiting times and lower congestion, outperforming conventional fixed-timing methods.

RESEARCH & PUBLICATIONS

Quantum Enhanced Threat Detection with Grover's Algorithm

September 2024 – Present

- Optimized Grover's Algorithm for feature selection to search for variables that predict anomalous network behavior
- Accepted and Published:** Cheng, Tiffany, Shoshana Sugerman, **Colin Farley**, and Brian Robert Callahan. 2025. "Quantum preprocessing for Internet of Things edge computing security devices." *Proceedings of the 1st IEEE Conference on Secure and Trustworthy CyberInfrastructure for IoT and Microelectronics (SaTC 2025)*. Wright State University. Dayton, OH. Feb. 25–27, 2025.

TECHNICAL SKILLS

Languages: C++, Python, C, Go, SQL, Java, JavaScript, HTML/CSS, Dart, MIPS, LaTeX

Tools & Frameworks: Git, Linux, Next.js, Qiskit, Flutter, PowerBI

Quantum: IBM Quantum Excellence Badge, Qiskit, Grover's Algorithm, Simulated Annealing

Relevant Courses: AI & ML, Quantum Systems, Network Programming, Databases, Data Structures, Algorithms